

Introduction to Clustering

Clustering is a type of unsupervised learning algorithm. Its goal is to look at a number of data points and automatically find points that are related or similar to each other, grouping them into "clusters."

Supervised vs. Unsupervised Learning

To understand clustering, it helps to contrast it with the supervised learning methods (like logistic regression) you learned previously.

Feature	Supervised Learning	Unsupervised Learning
Data	Includes input features (x) and target labels (y) .	Includes only input features (x) . No target labels are provided.
Visualization	Data points are distinct (e.g., x 's and y 's).	Data points look the same (e.g., all dots).
Goal	Find a decision boundary to predict the "right answer" (y).	Find "interesting structure" or patterns within the data.
Guidance	We tell the algorithm what the correct output is.	We ask the algorithm to figure it out on its own.

- **In the plot above:** You can see how an unsupervised algorithm looks at raw data (without labels) and identifies distinct groups (clusters) based on the proximity of the points to one another.

Real-World Applications of Clustering

Clustering is widely used across many industries to find hidden structures in data:

- **News Aggregation:** Grouping similar news articles together (e.g., grouping all stories about "Pandas" or a specific event) to present them as a single topic.
- **Market Segmentation:** Analyzing a database of customers to discover distinct groups.
 - *Example:* Deeplearning.ai discovered groups such as "Career Developers," "Skill Growers," and "AI Updaters" to better tailor content.
- **DNA Analysis:** Analyzing genetic expression data to group individuals who exhibit similar genetic traits.
- **Astronomy:** Analyzing astronomical data to group celestial bodies together to identify galaxies or coherent structures in space.